PromptLite-Seg: Lightweight Zero-Shot Prompted Segmentation on PASCAL VOC 2012

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Abstract

PromptLite-Seg studies zero-shot prompted segmentation on PASCAL VOC 2012. Given a weak prompt consisting of a center point and a bounding box, the goal is to segment the target object without training on the benchmark. The project implements a simple center-color baseline, a stronger SAM-free robust superpixel method, and a GPU SAM ViT-B comparison. It further introduces a prompt-perturbation robustness benchmark. On a 30-sample VOC 2012 subset, SAM reaches 0.8325 mean IoU with clean prompts, but drops to 0.6756 under moderate prompt noise. This shows that prompt quality remains a central bottleneck even for foundation segmentation models.

1 Introduction

Prompted segmentation has become a practical interface for general-purpose vision systems. Models such as the Segment Anything Model (SAM) demonstrate that segmentation can be controlled by points and boxes. However, large pretrained models may be expensive to run or difficult to reproduce in a lightweight course setting. This project therefore asks a complementary question: how far can a transparent, training-free prompted segmentation algorithm go on real benchmark images?

The task follows the course-design direction of zero-shot prompt segmentation. The benchmark is PASCAL VOC 2012. For each selected validation sample, the largest semantic object component is converted into a binary target mask. The prompt consists of a tight bounding box and an interior center point derived from the distance transform of the target mask.

2 Method

2.1 Prompt Generation

PASCAL VOC semantic masks provide foreground labels from 1 to 20 and background label 0. For each sample, PromptLite-Seg selects the largest connected foreground component as the target instance. The bounding box is the tight rectangle around this component. The point prompt is the pixel with maximum distance-transform value inside the component, which approximates a stable human center click.

2.2 Center-Color Baseline

The baseline estimates the target appearance from a small disk around the center point. Pixels inside the prompt box are segmented if their RGB distance to this median color prototype is below an adaptive Otsu threshold. Small components are removed, holes are filled, and the connected component nearest to the prompt point is retained.

2.3 Robust Superpixel Prompt Segmentation

The robust method first runs the center-color baseline to obtain a conservative foreground seed. It then oversegments the image into SLIC superpixels. Each superpixel is represented by mean Lab color and normalized spatial coordinates. Foreground prototypes are estimated from superpixels intersecting the center-point disk, while background prototypes are estimated from superpixels outside the prompt box and along the box border.

Each superpixel receives a foreground score based on its relative distance to foreground and background prototypes, combined with a weak spatial prior centered at the prompt point. The final mask is the union of the conservative color seed and superpixel consensus across slightly perturbed boxes, followed by morphology and point-connected component selection.

3 Experimental Setup

The experiment uses a compact subset of PASCAL VOC 2012 validation samples from the Hugging Face mirror `nateraw/pascal-voc-2012`. The subset contains 30 samples and 12 object classes. The evaluation reports Intersection over Union (IoU) and Dice score:

$$\mathrm{IoU} = \frac{|P \cap G|}{|P \cup G|}, \qquad \mathrm{Dice} = \frac{2|P \cap G|}{|P| + |G|}.$$

The reproducible commands are:

```
python scripts/download_voc_subset.py --count 30
python scripts/run_experiment.py --max-samples 30
python scripts/analyze_results.py
.\.venv-sam\Scripts\python.exe scripts/run_robustness_experiment.py ^
  --max-samples 30 --trials 2 --include-sam --device cuda
```

4 Results

Table 1: Aggregate results on the 30-sample VOC 2012 subset.

Method	Mean IoU	Std IoU	Mean Dice	Std Dice
Center-color baseline	0.5468	0.2323	0.6754	0.2122
Robust superpixel	0.6031	0.2111	0.7277	0.1895
SAM ViT-B point+box	0.8325	0.1484	0.9002	0.1044

The robust method improves mean IoU by 0.0562 absolute points and mean Dice by 0.0523. At the sample level, it wins on 23 samples, ties on 5 samples, and loses on only 2 samples relative to the center-color baseline.

The optional GPU SAM ViT-B comparison reaches 0.8325 mean IoU and 0.9002 mean Dice. This confirms that the prompts generated by the benchmark pipeline are informative enough for a modern foundation segmentation model, while the robust superpixel method remains useful as a transparent low-resource baseline.

4.1 Prompt Robustness

To evaluate prompt sensitivity, the robustness benchmark perturbs both the center point and the bounding box. Mild perturbations shift the point and box by roughly 5–6% of object size; moderate perturbations use roughly 10–12%. Each non-clean severity is evaluated with two deterministic trials per sample.

Table 2: Robustness under prompt perturbation.

Severity	Method	Mean IoU	Mean Dice
clean	robust superpixel	0.6031	0.7277
mild	robust superpixel	0.5579	0.6927
moderate	robust superpixel	0.4568	0.5967
clean	SAM ViT-B noisy prompt	0.8325	0.9002
mild	SAM ViT-B noisy prompt	0.7731	0.8537
moderate	SAM ViT-B noisy prompt	0.6756	0.7795

The robustness benchmark reveals the main limitation of prompt segmentation. SAM has much higher absolute accuracy, but it still loses about 0.157 IoU under moderate prompt noise. A naive lightweight prompt repair strategy does not improve SAM: it reduces moderate-noise SAM IoU from 0.6756 to 0.5745. SAM's own score-based selection also underperforms the original noisy prompt at 0.6118. This negative result suggests that prompt repair should preserve uncertainty instead of collapsing a noisy prompt into a single mask-derived point and box.

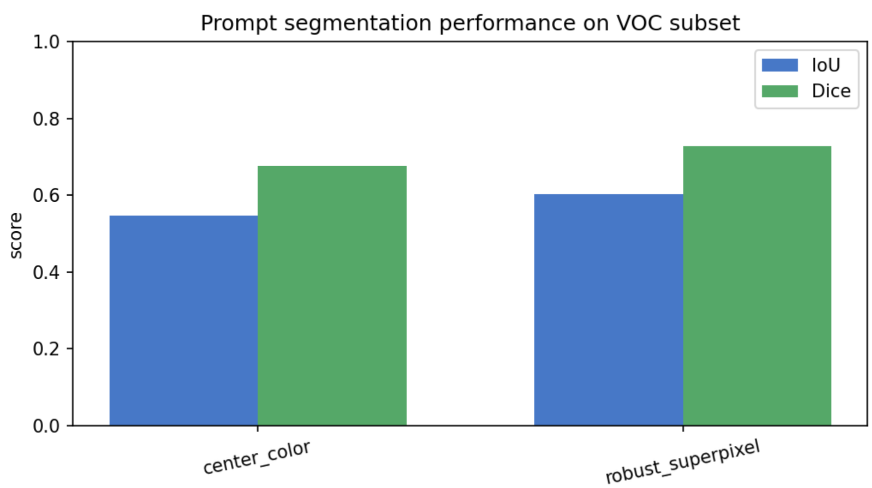

Figure 1: Aggregate IoU and Dice comparison.

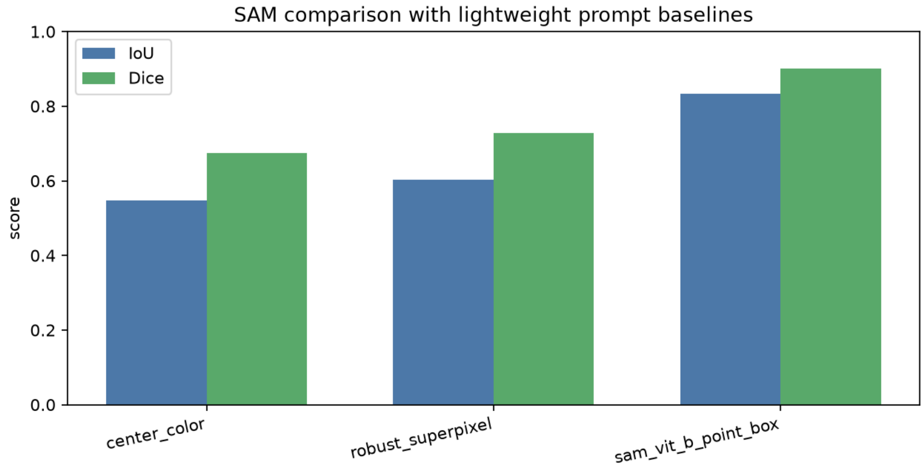

Figure 2: Comparison between lightweight prompt baselines and GPU SAM ViT-B.

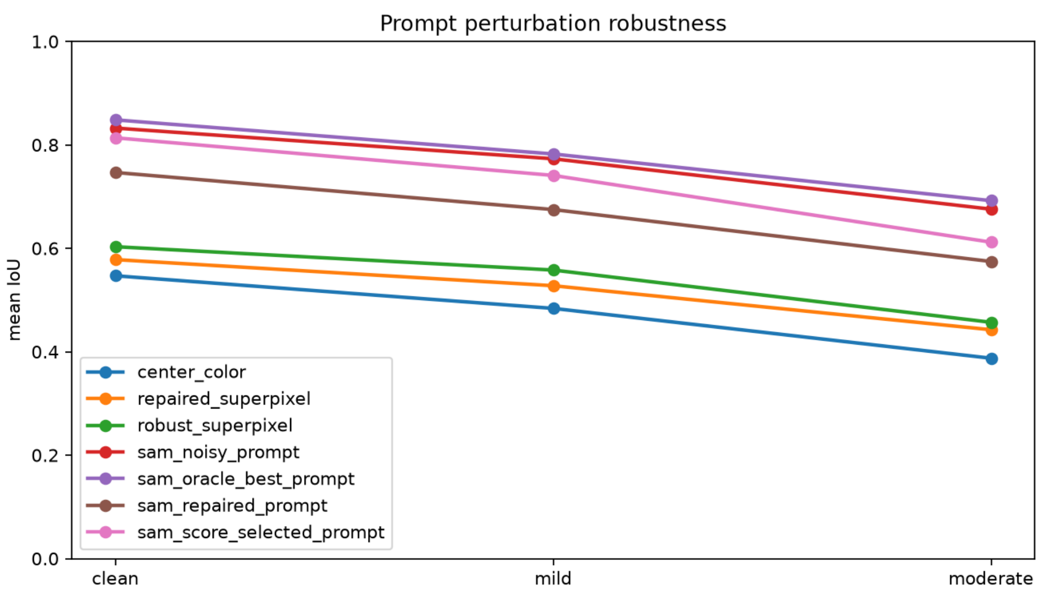

Figure 3: Prompt perturbation robustness under clean, mild, and moderate prompt noise.

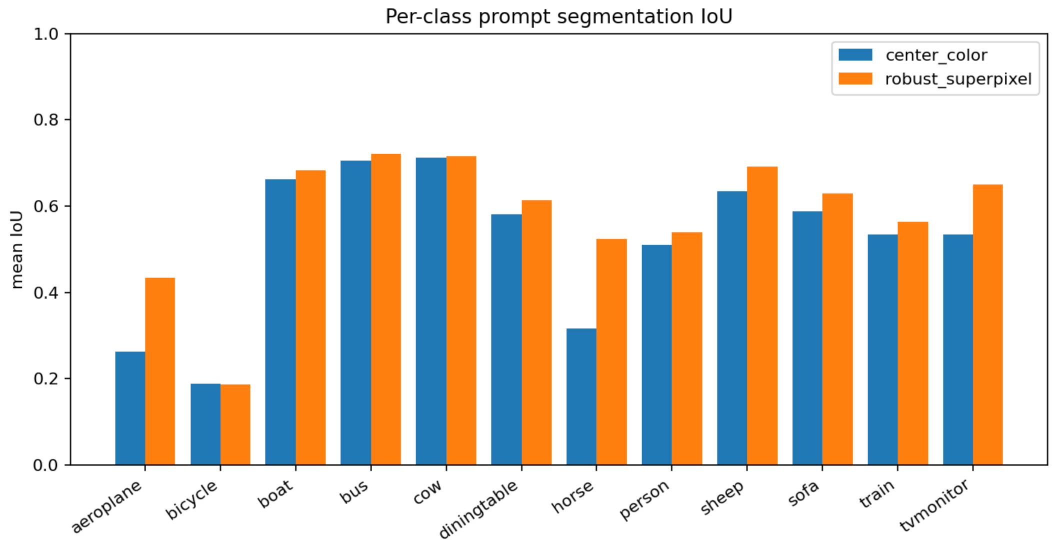

Figure 4: Per-class mean IoU. The strongest gains appear on horse, tvmonitor, and aeroplane examples.

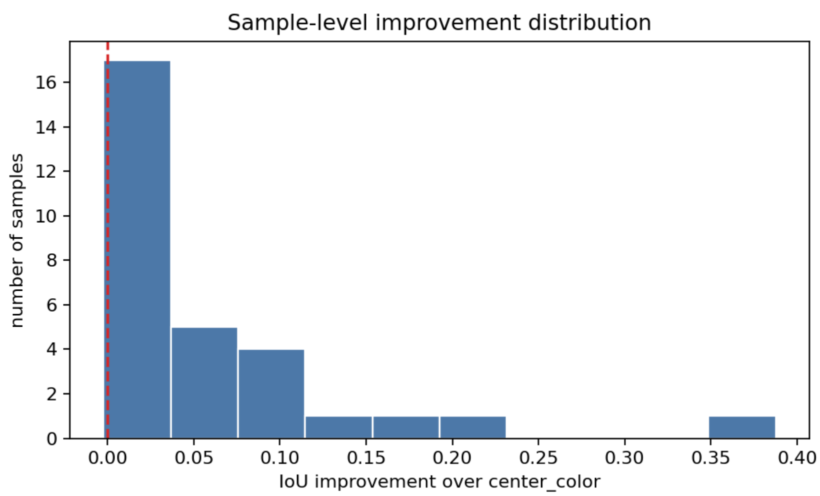

Figure 5: Distribution of sample-level IoU improvement over the center-color baseline.

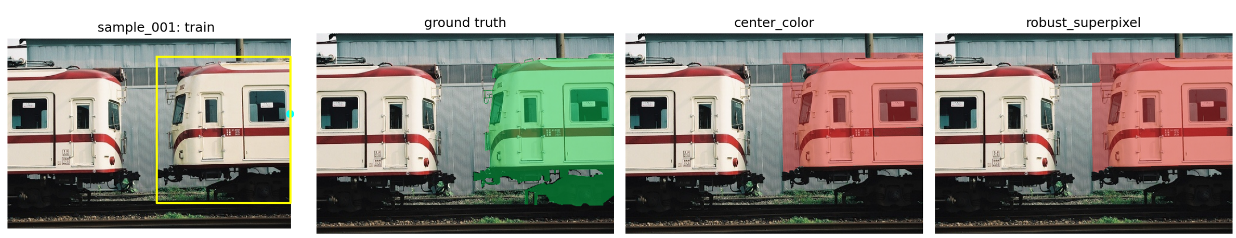

Figure 6: Qualitative example showing prompt, ground truth, center-color prediction, and robust superpixel prediction.

5 Analysis

The center-color baseline works when the target has a compact appearance and contrasts clearly with the local background. It fails when the object has multiple colors, when the center click lands on a non-representative part, or when background regions share similar colors.

The robust method improves the prompt interface in three ways. First, superpixels better respect local image boundaries than raw pixel thresholding. Second, background evidence from the box boundary and outside-box region suppresses leakage to nearby clutter. Third, box perturbation makes the prediction less sensitive to a single exact prompt.

The hardest cases include train, bicycle, diningtable, and boat images. Trains and dining tables often contain repetitive interior structures, bicycles contain thin parts that are easy to miss, and boats may include strong background colors inside the prompt box. The robustness experiment further shows that prompt noise is not a small implementation detail: it changes the ranking and reliability of methods.

6 SAM Comparison and Lightweight Baseline

PromptLite-Seg is intentionally designed so that its main baseline remains SAM-free, CPU-friendly, deterministic, and easy to inspect. With GPU resources available, the project also evaluates SAM ViT-B using the same point-and-box prompts. SAM achieves much higher absolute accuracy, especially on objects with complex boundaries, while the lightweight method provides an interpretable baseline and a useful diagnostic fallback.

The deeper finding is that a stronger model is not automatically robust to prompt uncertainty. The oracle best-of-two prompt selection only slightly improves SAM, while score-based prompt selection is worse than simply keeping the noisy prompt. This exposes a concrete future direction: robust prompted segmentation should model prompt uncertainty explicitly, for example by maintaining multiple plausible prompts or by learning a calibrated prompt-quality estimator.

7 Conclusion

This project implements and evaluates a zero-shot prompted segmentation pipeline on PASCAL VOC 2012. It satisfies the course requirement of running an algorithm on a benchmark, and it adds analysis beyond aggregate scores through per-class, success/failure, SAM comparison, and prompt-robustness experiments. The main research conclusion is that prompt quality is a measurable bottleneck: SAM is strong under clean prompts, but prompt noise causes substantial degradation, and naive repair is not sufficient.

References

[1] Mark Everingham, Luc Van Gool, Christopher K. I. Williams, John Winn, and Andrew Zisserman. The PASCAL Visual Object Classes Challenge 2012. `https://www.robots.ox.ac.uk/~vgg/projects/pascal/VOC/voc2012/`

[2] Alexander Kirillov et al. Segment Anything. ICCV 2023. `https://arxiv.org/abs/2304.02643`

[3] Radhakrishna Achanta et al. SLIC Superpixels Compared to State-of-the-Art Superpixel Methods. IEEE TPAMI 2012.